Pre-Analysis Plan

#ManyDaughters Project

Research Team

Research Team :

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Plan for answering the hypothesis H1:

Hypothesis: *Having daughters leads to stronger support for gender equality (as measured by attitudes).*

1) Variables:

Dependent variable: Attitudinal measure for the support for gender equality (SGE)

- Sum index of eight items from the year 2018 of the SOEP-core questionnaire: plh0300_v2, plh0357 plh0358 plh0302_v2 plh0298_v2 plh0308_v2 plh0359 plh0360 (pl.dta dataset). The items plh0300_v2 plh0302_v2 plh0298_v2 will be used inverted in the sum index so that higher scores equal more stronger support for gender equality on all items.

  (Since we already worked with these items in the past, we know that the scale has an acceptable level of reliability (Cronbach's α was above 0.7).)

Key independent variable: Daughter vs. son as first child (DvS1)

- Dummy variable: 0 – "Son as the only child born since 2014" vs. 1 – "Daughter as the only child born since 2014" based on information of the variables kidgeb01 kidsex01, kidgeb02, and kidsex02 (biobirth.dta dataset).

Table 1: Control variables

Name	**Scale**	**Variable name in the SOEP**
Sex	Dummy; Recoded into 0 – "Female", 1 – "Male"	ppath/sex
Age	Metric; in years	2018- gebjahr biobirth/gebjahr
Age²	Metric; in years squared	2018- gebjahr biobirth/gebjahr
Socialization in East/ West Germany	Dummy; 0 – "socialized/living in federal state formerly part of Federal Republic of Germany" 1 – "socialized/living in federal state formerly part of German Democratic Republic"	Recoded from: loc1989, birthregion_ew sampreg ppathl/loc1989 ppfad/birthregion_ew ppathl/sampreg
Level of education (recoded version of CASMIN or ISCED classification)	Categorical; Recoded into three categories: 0 – "Low educational level", 1 – "Medium educational level", 2 – "High educational level"	Recoded from pgen/pgcasmin
Household income	Categorical; Net household income from last month recoded into deciles	Recoded from hl/hlc0005_h
Employment status and occupational position	Categorical; Condensed categorical variables. 1 – "Unemployed", 2 - "In education / apprenticeship", 3- "Pensioner", 4 – "In maternity/parental leave", 5 – "Worker" (part time), 6 – "Worker (full-time)", 7 – "Self-employed/freelancer (part-time)", 8 – "Self-employed /freelancer (full-time)", 9 – "Salaried employee (part-time)", 10 – "Salaried employee (full-time)", 11 - "Civil servant (part-time)", 12 - "Civil servant (full-time)". If we run into overspecification problems with this variable (i.e. model not estimable) we reduce the number of categories by dropping the part-time/full-time split.	Recoded from pgemplst, pgstib and pl/plb0019_v2
Religious denomination	Categorical; 0 "No confession", 1 – "Catholic", 2 – "Protestant", 3 – "Muslim", 4 "Other confession"	Recoded version from pl/plh0258_h

Attendance of religious services	Categorical; Recoded into three categories 0 – "Never", 1 – "Less than once a month", 2 – "At least once a month"	Recoded from pl/pli0098_v2 (latest available information used)
Legal marital status	Dummy; 0 – "Neither married nor cohabitating" 1 – "Married or cohabitating"	Recoded from Pgfamstd
Having a sister	Dummy; 0 – "No sister", 1 – "At least one sister"	Both variables generated from lb1092_v1 lb1092_v2 lb1093_v1 lb1093_v2 lb1094_v1 lb1094_v2 lb1095_v1 lb1095_v2 lb1096_v1 lb1096_v2 lb1097_v1 lb1097_v2 lb1098_v1 lb1098_v2 lb1099_v1 lb1099_v2 lb1100_v1 lb1100_v2 lb1101_v1 in biol
Having a brother	Dummy; 0 – "No brother", 1 – "At least one brother"	
Multi-generational household	Dummy; 0 – "No", 1 – "Yes"	Recoded from hgen/hgtyp1hh
Big 5: Openness and agreeableness	Metric; Sumscores for openness and agreeableness separately for the latest available survey year where the women was at the time of the interview no longer than 3 months pregnant (legal restriction for the prenatal identification of gender, checked with birthyear and -month based on biobirth.dta)	Openness: plh0255 plh0215 plh0220 plh0225 Agreeableness: plh0214 plh0217 plh0224
Single parent	Dummy variable; 0 – "No", 1 – "Yes".	Recoded from hgen/hgtyp1hh
Survey year	Categorical; based on syear	syear
Sample	Categorical, based on psample	psample

2) Inclusion-/exclusion criteria:

1. We use the sample of 2018 with additional information based on the birth of children using biobirth-data. We restrict the sample to adult parents with a minimum age of 18 years who just received their first child within a time frame of 5 years (2014-2018) before gender-attitudes have been surveyed in 2018. Parents in our sample were not allowed to have additional children until 2018.
2. Dealing with missing values on control variables: We exclude all cases with a missing value on at least one of the considered variables (listwise deletion), provided that the loss of cases is below 10%. If the loss of cases exceeds 10%, the variables with the most missing values are subsequently excluded until the loss of cases is below 10%.

3) Exact analysis:

We use a linear OLS regression model with clustered standard errors at the household level. The model is specified as follows.

Formula 1: OLS model for hypothesis 1

$$SGE_i = \beta_0 + \beta_1 DvS1_i + \beta_k X_{ki} + u_i$$

Notation: SGE = support for gender equality, DvS1 = Daughter vs. Son, X_{ki} = Term including the information of the k control variables (included as metric and categorical/dummy as indicated in Table 1)

4) Interpretation of results

We reject the null hypothesis (i.e. H0: having daughters does not lead to more support for gender equality) if there is a **positive regression coefficient for β_1** (i.e. the coefficient for DvS1 indicating higher support for gender equality amongst individual parents of a daughter compared to individual parents of a son) **and** a **p-value for a two-sided test below 0.05** for this coefficient β_1 being zero (5 percent significance level for the coefficient). A rejection of the null hypothesis will be interpreted as support for the hypothesis H1.

Plan for answering the hypothesis H2:

Hypothesis: *Having daughters leads to more left-leaning political preferences.*

1) Variables:

Dependent variable: political left-right position (LR)

To measure left-right political preferences we use the left-right-scale (pl/plh0004). This variable is surveyed in 4 waves (2005, 2009, 2014 and 2019) and measured on a 11 scale from 0 (completely left) to 10 (completely right) and used as it is after removing missing values.

Key independent variables: First child (FC) and daughter vs son (DvS2)

We generate the following dummy variables

1. First-child-dummy: 0 "No child", 1 "A first child was born still being the only child" (FC)
2. Time-constant sex of first child: 0 "Male (son)", 1 "Female (daughter)" (DvS2)

These two dummy variables are going to be constructed using the information of the variables kidgeb01 kidsex01, kidgeb02, and kidsex02 from the biobirth-data set. The variable kidgeb01 indicates the exact year in which the first child is born. We are going to assign zeros to the dummy variables if the observation time point lies before the event of the birth of the first son/the first daughter. We assign ones if the observation time point is after the birth of the first son/first daughter and no second child was born yet (identifiable with kidgeb02). The sex of the corresponding first child is used as a time-constant variable over the relevant observational period (i.e. already contains the sex of the first child before birth).

Later on, we merge our constructed dummy variables onto the 4-wave-sample including the years 2005, 2009, 2014 and 2019.

Table 2: Control variables

Name	Scale & Recoding	Variable name in the SOEP
Age	Metric; in years	2018- gebjahr biobirth/gebjahr
Age²	Metric; in years squared	2018- gebjahr biobirth/gebjahr
Level of education (recoded version of CASMIN or ISCED classification)	Categorical; Recoded into three categories: 0 – "Low educational level", 1 – "Medium educational level", 2 – "High educational level"	Recoded from pgen/pgcasmin
Household income	Categorical; Net household income from last month recoded into deciles	Recoded from hl/hlc0005_h
Employment status and occupational position	Categorical; Condensed categorical variables. 1 – "Unemployed", 2 - "In education / apprenticeship", 3- "Pensioner", 4 – "In maternity/parental leave", 5 – "Worker" (part time), 6 – "Worker (full-time)", 7 – "Self-employed/freelancer (part-time)", 8 – "Self-employed /freelancer (full-time)", 9 – "Salaried employee (part-time)", 10 – "Salaried employee (full-time)", 11 - "Civil servant (part-time)", 12 - "Civil servant (full-time)". If we run into overspecification problems with this variable (i.e. model not estimable) we reduce the number of categories by dropping the part-time/full-time split.	Recoded from Pgemplst, pgstib and pl/plb0019_v2
Attendance of religious services	Categorical; 0 "No confession", 1 – "Catholic", 2 – "Protestant", 3 – "Muslim", 4 "Other confession"	Recoded version from pl/plh0258_h (latest available information used)
Single parent	Dummy; 0 – "No", 1 – "Yes".	Recoded from hgen/hgtyp1hh

2) Inclusion-/exclusion criteria:

1. Our sample is restricted to adults with a minimum age of 18 years. The sample is restricted to waves in which the left-right scale is surveyed. These are four waves: 2005, 2009, 2014 and 2019.
2. We only include respondents who initially were no parents, but who become parent of a first child somewhere between 2005 and 2019. Following the definition of the first-daughter/first-son dummy variables we keep those observations from these respondents in which they either had no children or one child. As soon as second child is born we drop observations following the event of the birth of the second child. If respondents directly "jump" from having zero children to having 2 or more children we drop these respondents completely from the sample.
3. We keep observations around the event of the birth of the first child. We keep observations from 5 years before the birth of the first child and 5 years after the birth of the first child. E.g. if a child is born in 2007 we use only the waves 2005 and 2009 since they are within the -5/+5-time window.
4. Dealing with missing values on control variables: We exclude all cases with a missing value on at least one of the considered variables (listwise deletion), provided that the loss of cases is below 10%. If the loss of cases exceeds 10%, the variables with the most missing values are subsequently excluded until the loss of cases is below 10%.

3) Exact analysis:

We use fixed effects (FE) regression models with clustered standard errors at the household level. The model is specified as follows.

Formula 2: Fixed effects model for hypothesis 2

$$(LR_{it} - \overline{LR_{\iota}}) = \beta_1(FC_{it} - \overline{FC_{\iota}}) + \beta_2(FC_{it} * DvS2_{it} - \overline{FC_{\iota} * DvS2_{\iota}}) + \beta_k(X_{kit} - \overline{X_{k\iota}}) + u_{it}$$

*Note: Practically, the FC*DvS2 interaction is simply included as such in the fixed effects model to be calculated with "xtreg, fe" in Stata. Technically, the model demeanes the interaction as formulized above.*

Notation: LR = political left-right position, DvS2 = Daughter vs. Son (Daughter=1), X_k = Term including information of the k control variables (included as metric and categorical/dummy as indicated in Table 2)

4) Interpretation of results

We reject the null hypothesis (i.e. H0: having daughters does not lead to having more left-leaning political preferences) if there is a **negative regression coefficient for β_2** (i.e. the coefficient for the FC×DvS2 interaction term indicating more left-leaning political positions after birth amongst individual parents who received a daughter compared to individual parents who received a son) **and** a **p-value for a two-sided test below 0.05** for this coefficient β_2 being zero (5 percent significance level for the coefficient). A rejection of the null hypothesis will be interpreted as support for the hypothesis H2.

Plan for answering the hypothesis H3:

Hypothesis: *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes)*

1) Variables:

Dependent variable: Division of work for measuring gender equality in the household (D)

We measure gender equality in the household through the division of work in the relationship. We construct an index which takes both partners' hours per day being spend for paid work and their hours being spend for care work into account. By using the partner identifier "parid" we are able to identify couples in the SOEP data. This makes it possible to construct the following couple-specific index for the division of work in the relationship.

The index is defined as follows:

$$D = \left| \frac{paid\ work_A}{(paid\ work_A + \ paid\ work_B)} - \frac{care\ work_A}{(care\ work_A + \ care\ work_B)} \right|$$

This index ranges from 0 (egalitarian division of work) to 1 (complete separation of tasks).

Hours of paid work pl/pli0038_h
Hours of care work pl/pli0043_h
Both variables are surveyed annually from 1984 until 2022

Key independent variables: First child (FC) and daughter vs son (DvS2)

We generate the following dummy variables

1. First-child-dummy: 0 "No child", 1 "A first child was born still being the only child" (FC)
2. Time-constant sex of first child: 0 "Male (son)", 1 "Female (daughter)" (DvS2)

These two dummy variables are going to be constructed using the information of the variables kidgeb01 kidsex01, kidgeb02, and kidsex02 from the biobirth-data set. The variable kidgeb01 indicates the exact year in which the first child is born. We are going to assign zeros to the dummy variables if the observation time point lies before the event of the birth of the first son/the first daughter. We assign ones if the observation time point is after the birth of the first son/first daughter and no second child was born yet (identifiable with kidgeb02). The sex of the corresponding first child is used as a time-constant variable over the relevant observational period (i.e. already contains the sex of the first child before birth).

Table 3: Control variables

Name	Scale & Recoding	Source in the SOEP
Age	Metric; in years	2018- gebjahr ppath/gebjahr
Age²	Metric; in years squared	2018- gebjahr ppath/gebjahr
Level of education (recoded version of CASMIN or ISCED classification)	Categorical; Recoded into three categories: 0 – "Low educational level", 1 – "Medium educational level", 2 – "High educational level"	Recoded from pgen/pgcasmin
Household income	Categorical; Net household income from last month recoded into deciles	Recoded from hl/hlc0005_h
Employment status and occupational position	Categorical; Condensed categorical variables. 1 – "Unemployed", 2 - "In education / apprenticeship", 3- "Pensioner", 4 – "In maternity/parental leave", 5 – "Worker" (part time), 6 – "Worker (full-time)", 7 – "Self-employed/freelancer (part-time)", 8 – "Self-employed	Recoded from pgemplst, pgstib and pl/plb0019_v2

	/freelancer (full-time)", 9 – "Salaried employee (part-time)", 10 – "Salaried employee (full-time)", 11 - "Civil servant (part-time)", 12 - "Civil servant (full-time)". If we run into overspecification problems (i.e. model not estimable) with this variable we reduce the number of categories by dropping the part-time/full-time split.	
Attendance of religious services	Categorical; 0 "No confession", 1 – "Catholic", 2 – "Protestant", 3 – "Muslim", 4 "Other confession"	Recoded version from pl/plh0258_h (latest available information used)

2) Inclusion-/exclusion criteria:

1. Our sample is restricted to adults with a minimum age of 18 years who are in a relationship and stay in it within the observational window.
2. We only include individuals who initially were no parents, but who become parent of a first child at some point in the SOEP. Following the definition of the first-daughter/first-son dummy variables we keep those observations from these respondents in which they either had no children or one child. As soon as a second child is born we drop observations following the event of the birth of the second child. If respondents directly "jump" from having zero children to having 2 or more children we drop these respondents completely from the sample.
3. We keep observations in the sample around the event of the birth of the first child. We keep observations from 5 years before the birth of the first child and 5 years after the birth of the first child.
4. Since the division of work index uses information of both partners, the sample is restricted to parents in couples living together in one household. Furthermore, the sample is restricted to those couples in the SOEP in which both partners provided information on their amount of paid work and care work.
5. Dealing with missing values on control variables: We exclude all cases with a missing value on at least one of the considered variables (listwise deletion), provided that the loss of cases is below 10%. If the loss of cases exceeds 10%, the variables with the most missing values are subsequently excluded until the loss of cases is below 10%.

3) Exact analysis:

We use fixed effects (FE) regression models with clustered standard errors at the household level. The model is specified as follows.

Formula 3: Fixed effects model for hypothesis 3

$$(D_{it} - \overline{D_{\iota}}) = \beta_1(FC_{it} - \overline{FC_{\iota}}) + \beta_2(FC_{it} * DvS2_{it} - \overline{FC_{\iota} * DvS2_{\iota}}) + \beta_k(X_{kit} - \overline{X_{k\iota}}) + u_{it}$$

*Note: Practically, the FC*DvS2 interaction is simply included as such in the fixed effects model to be calculated with "xtreg, fe" in Stata. Technically, the model demeanes the interaction as formulized above.*

Notation: D = Division of work, FC = First Child born, FvM = First Child Born is female (time constant, Ref.: male); X_k = Vector for the k control variables (included as metric and categorical/dummy as indicated in Table 3)

4) Interpretation of results

We reject the null hypothesis (i.e. H0: having daughters does not lead to more gender quality in the household) if there is a **negative regression coefficient for β_2** (i.e. the coefficient for the FC×DvS2 interaction term indicating more gender equality in the household among parents who received a daughter compared to parents who received a son) **and** a **p-value for a two-sided test below 0.05** for this coefficient β_2 being zero (5 percent significance level for the coefficient). A rejection of the null hypothesis will be interpreted as support for the hypothesis H3.

Plan for answering the hypothesis H4

Hypothesis: *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes)*

1) Variables:

Dependent variable: Attitudinal measure for the support for gender equality (SGE)

- Sum index of eight items from the year 2018 of the SOEP-core questionnaire: plh0300_v2 plh0357 plh0358 plh0302_v2 plh0298_v2 plh0308_v2 plh0359 plh0360 (pl data.dta dataset). The items plh0300_v2 plh0302_v2 plh0298_v2 will be used inverted in the sum index so that higher scores equal more stronger support for gender equality on all items.

  (Since we already worked with these items in the past, we know that the scale has an acceptable level of reliability (Cronbach's α was above 0.7).)

Key independent variables: Daughter vs. son as first child (DvS1) and respondent's sex (Sex)

- dummy variable: 0 – "Son as the only child born since 2014" vs. 1 – "Daughter as the only child born since 2014" based on information of the variables kidgeb01 kidsex01, kidgeb02, and kidsex02 (biobirth.dta dataset). (DvS1)
- To test how sex moderates the daughter effect we use an interaction between the daughter-dummy (DvS1) and the sex of the respondent (recoded in to 0 –"Female", 1 – "Male", biobirth/sex). (Sex)

Table 4: Control variables

Name	Scale & Recoding	Source in the SOEP
Age	Metric; in years	2018- gebjahr ppath/gebjahr
Age²	Metric; in years squared	2018- gebjahr ppath/gebjahr
Socialization in East/ West Germany	Dummy; 0 – "socialized/living in federal state formerly part of Federal Republic of Germany" 1 – "socialized/living in federal state formerly part of German Democratic Republic"	Recoded from: loc1989, birthregion_ew sampreg ppathl/loc1989 ppfad/birthregion_ew ppathl/sampreg
Level of education (recoded version of CASMIN or ISCED classification)	Categorical; Recoded into three categories: 0 – "Low educational level", 1 – "Medium educational level", 2 – "High educational level"	Recoded from pgen/pgcasmin
Household income	Categorical; Net household income from last month recoded into deciles	Recoded from hl/hlc0005_h
Employment status and occupational position	Categorical; Condensed categorical variables. 1 – "Unemployed", 2 - "In education / apprenticeship", 3- "Pensioner", 4 – "In maternity/parental leave", 5 – "Worker" (part time), 6 – "Worker (full-time)", 7 – "Self-employed/freelancer (part-time)", 8 – "Self-employed /freelancer (full-time)", 9 – "Salaried employee (part-time)", 10 – "Salaried employee (full-time)", 11 - "Civil servant (part-time)", 12 - "Civil servant (full-time)". If we run into overspecification problems with this variable (i.e. model not estimable) we reduce the number of categories by dropping the part-time/full-time split.	Recoded from pgemplst, pgstib and pl/plb0019_v2
Religious denomination	Categorical; 0 "No confession", 1 – "Catholic", 2 – "Protestant", 3 – "Muslim", 4 "Other confession"	Recoded version from pl/plh0258_h

Attendance of religious services	Categorical; Recoded into three categories 0 – "Never", 1 – "Less than once a month", 2 – "At least once a month"	Recoded from pl/pli0098_v2 (latest available information used)
Legal marital status	Dummy; 0 – "Neither married nor cohabitating" 1 – "Married or cohabitating"	Recoded from pgfamstd
Having a sister	Dummy; 0 – "No sister", 1 – "At least one sister"	Both variables generated from lb1092_v1 lb1092_v2 lb1093_v1 lb1093_v2 lb1094_v1 lb1094_v2 lb1095_v1 lb1095_v2 lb1096_v1 lb1096_v2 lb1097_v1 lb1097_v2 lb1098_v1 lb1098_v2 lb1099_v1 lb1099_v2 lb1100_v1 lb1100_v2 lb1101_v1 in biol
Having a brother	Dummy; 0 – "No brother", 1 – "At least one brother"	
Multi-generational household	Dummy; 0 – "No", 1 – "Yes"	Recoded from hgen/hgtyp1hh
Big 5: Openness and agreeableness	Metric; Sumscores for openness and agreeableness separately for the latest available survey year where the women was at the time of the interview no longer than 3 months pregnant (legal restriction for the prenatal identification of gender, checked with birthyear and -month based on biobirth.dta)	Openness: plh0255 plh0215 plh0220 plh0225 Agreeableness: plh0214 plh0217 plh0224
Single parent	Dummy; 0 – "No", 1 – "Yes".	Recoded from hgen/hgtyp1hh
Survey year	Categorical; based on syear	syear
Sample	Categorical, based on psample	psample

2) Inclusion-/exclusion criteria:

1. We use the sample of 2018 with additional information based on the birth of children using biobirth-data. We restrict the sample to adult parents with a minimum age of 18 years who just received their first child within a time frame of 5 years (2014-2018) before gender-attitudes have been surveyed in 2018. Parents in our sample were not allowed to have additional children until 2018.
2. Dealing with missing values on control variables: We exclude all cases with a missing value on at least one of the considered variables (listwise deletion), provided that the loss of cases is below 10%. If the loss of cases exceeds 10%, the variables with the most missing values are subsequently excluded until the loss of cases is below 10%.

3) Exact analysis:

As for the first hypothesis, we use a linear OLS regression model with clustered standard errors at the household level. The model is specified as follows.

Formula 4: OLS model for hypothesis 4

$$SGE_i = \beta_0 + \beta_1 DvS1_i + \beta_2 Sex_i + \beta_3 (DvS1_i * Sex_i) + \beta_k X_{ki} + u_i$$

Notation: SGE = support for gender equality, DvS1 = Daughter vs. Son (1=daughter), Sex = Respondent's Sex (1=male); X_k = Term for the control variables (included as metric and categorical/dummy as indicated in Table 4)

4) Interpretation of results

We reject the null hypothesis (i.e. H0: having a daughter does not increase support for gender equality stronger for men (fathers) compared to women (mothers)) if there is a **positive regression coefficient for β_3** (i.e. the DvS1×Sex interaction term indicating higher support for gender equality among male parents of a daughter compared to female parents of a daughter (compared to having a son)) **and** a **p-value for a two-sided test below 0.05** for this coefficient β_3 being zero (5 percent significance level for the coefficient). A rejection of the null hypothesis will be interpreted as support for the hypothesis H4.